

Laparoscopic adhesiolysis versus open adhesiolysis in acute adhesive small bowel obstruction.

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Introduction:

Exploratory laparotomy is still the standard therapy for patients who need surgical intervention for adhesive small bowel obstruction (SBO). However, the use of laparoscopy in the management of adhesive SBO is still controversial. We aimed to detect the short-term outcomes between open and laparoscopic adhesiolysis for SBO.

Patients and Methods:

This is a retrospective study of patients with adhesive SBO who underwent either laparoscopic or open surgery from June 2019 to July 2022 at Ain Shams University Hospitals. Intraoperative and early post-operative outcomes were compared in the two groups.

Results:

A total of 89 patients with adhesive SBO were included in our study. Fifty-one cases underwent open adhesiolysis and 38 cases underwent laparoscopic adhesiolysis. Laparoscopic adhesiolysis is associated with a remarkable decrease in the operative time (71 min vs. 107 min, $P = 0.001$) and blood loss (50 ml vs. 120 ml, $P = 0.001$) in comparison to open adhesiolysis. In addition to that, those who underwent adhesiolysis by laparoscopy had a short hospital stay (2.4 days vs. 3.8 days, $P = 0.001$), early recovery (time to pass flatus 1.3 days vs. 2.8 days) and less post-operative complications (surgical site infection [SSI] 2.6% vs. 19.6%, $P = 0.001$). Moreover, open adhesiolysis is associated with a higher rate of early post-operative mortality. In addition to that, the incidence of iatrogenic injury was higher in the open group.

Conclusion:

Laparoscopic adhesiolysis is a safe and feasible approach for the management of SBO and has better short-term outcomes, especially if done by skilled surgeons in advanced laparoscopic techniques.

INTRODUCTION

There are numerous causes of small bowel obstruction (SBO) including incarcerated hernia, small bowel malignancy, peritoneal carcinomatosis, bezoars and gall bladder ileus. However, adhesive obstruction related to previous surgery accounts for 60%–75% of the cases.[1,2] According to some studies, about 2.5%–6% of the patients who undergo abdominopelvic surgery will be re-admitted for SBO within 4 years from the initial operation.[3] In addition to that, despite being as low as 3%, the mortality rate of SBO reaches 7%–14% in elderly patients.[4,5]

Back in history, in the 16th century BC, the famous Ebers papyrus was the first documentation of SBO. Moreover, ancient Greece performed laparotomies as management for intestinal ‘knots’ with or without enterotomy to empty the digestive content.[6] In recent years, according to Bologna guidelines, exploratory laparotomy is still the standard therapy for patients who need surgical intervention for adhesive SBO.[7] Nevertheless, the use of laparoscopy in the management of adhesive SBO is still controversial.

On the one hand, some studies reported a dramatic decrease in the length of hospital stay and the overall incidence of complications associated with the laparoscopic approach for small bowel adhesiolysis.[8,9] On the other hand, others claimed the presence of selection bias in these studies for the less severe cases to be managed with laparoscopy.[7,10] Furthermore, the laparoscopic approach is associated with a higher risk of iatrogenic bowel injury.[11] In addition to that, according to a survey from the UK, only

50%–60% of the surgeons will prefer using the laparoscopic approach in adhesiolysis.[12]

We aimed to compare the overall incidence of complications and hospital stay for open versus laparoscopic adhesiolysis for SBO.

PATIENTS AND METHODS

Data collection

This study has been approved by the Ethics Committee of Ain Shams University Hospital. This is a retrospective comparative study of patients with adhesive SBO who underwent either laparoscopic or open surgery from June 2019 to July 2022 at Ain Shams University Hospitals (Ain Shams Specialized Hospital and El Demerdash Hospital) a tertiary hospital that serves about 3 million people.

Patients with SBO were identified from electronic patient records of admission and electronic patient records of the operating room. Patient medical history, clinical signs and findings before surgery and other perioperative and post-operative outcomes were obtained from patient medical records. Patient laboratory results and computed tomography (CT) findings were obtained from electronic medical records. Those who could not be reached after discharge was defined as ‘lost to follow-up’.

Baseline characteristics, pre-operative data and intraoperative and post-operative outcomes were compared between patients who underwent open adhesiolysis and lap adhesiolysis. Patients who underwent conversion from laparoscopic to open surgery were included in the open group. This data included patients’ age, sex, body mass index (BMI), American Society of Anesthesiologists score (ASA), number of previous operations, presence of comorbidities, CT finding, laboratory results, duration of symptoms before admission, duration of the surgery, blood loss, cause of SBO, presence of iatrogenic injury, time to pass flatus, length of hospital stay, post-operative complications and early post-operative mortality.

Operative technique

The choice of surgical approach was made according to the surgeon’s decision. Each surgeon selected the open or laparoscopic approach according to their criteria and laparoscopic skills. Haemodynamic instability, clearly hostile abdomen and patients with medical contraindications for pneumoperitoneum were considered contraindications for the laparoscopic approach and were included in the open group.

Regarding the laparoscopic technique, the patient is catheterised and positioned in a supine manner with the left arm adducted and the legs adducted. A table belt is placed over the patient’s chest securing the position. Insufflation of the abdomen starts using a Veress needle at the ‘palmer’s point’. Then, an optical viewport is inserted under direct visualisation at the left upper quadrant (LUQ) to achieve safe abdominal entry. After insufflating the peritoneal cavity, diagnostic laparoscopy is performed identifying a safe free area to insert the first working port at the left lumbar area. When necessary, we use the Ligasure® to free the omental adhesions from the anterior abdominal wall making it clear to safely insert the second working port at the suprapubic area, after which the patient is positioned in a steep Trendelenburg position with the right shoulder up. Identifying the adhesive obstructive band by gentle counting of the small bowel starts retrograde from the ileocecal junction till reaching the site of obstruction using intestinal graspers. Adhesiolysis using cold scissors or Ligasure when necessary was performed making sure that the distal bowel is then refilled and counting the bowel again till reaching proximally at the proximal jejunum.

Turning to the open technique, a midline incision was used. Whenever possible, the whole small intestine was freed from adhesions. The bowel was decompressed through a nasogastric tube.

Outcomes

Our outcome of interest was to compare the operative details (operative time, blood loss, cause of SBO and iatrogenic injury) and post-operative outcomes (time to pass flatus, length of hospital stay, 30-day post-operative mortality and the overall incidence of complications) between open and laparoscopic

approaches for adhesive SBO.

Statistical analysis

A descriptive analysis was performed. Categorical variables are expressed as counts and percentiles. Continuous variables were reported as mean and standard deviation. Chi-square and Fisher's exact tests were used for categorical variables. t-tests and Wilcoxon rank-sum tests were used for continuous variables. Data were analysed using SPSS software package version 13.0 (SPSS Inc., Chicago, IL, USA). $P < 0.05$ was considered statistically significant.

RESULTS

During this period, 213 cases of SBO were operated on in our hospital. One hundred and six cases were excluded from our study: 71 cases with an incarcerated hernia, 12 cases with primary or secondary malignancy, 8 cases with mesenteric torsion, 7 cases with gallstone ileus, 5 cases with a bezoar and 3 cases with an internal hernia. In addition to that, we lost follow-up for 11 patients who underwent surgery for adhesive SBO, and 7 cases who underwent adhesiolysis with bowel resection due to ischaemic bowel were excluded from the study. A total of 89 cases with adhesive SBO were included in our study. Fifty-one cases underwent open adhesiolysis and 38 underwent laparoscopic adhesiolysis. Three cases started laparoscopically and converted into open; two cases needed dense adhesions and one case could not detect the cause of obstruction [Figure 1].

Flow diagram of the included cases

Baseline characteristics

To begin with the basic characteristics of the patients and pro-operative data, despite the presence of minor variability between the two groups, no statistically significant difference could be detected between those who underwent open and laparoscopic approaches. The mean age in the open group was 43.88 years (17.6) and 46.21 years (17.56) in the laparoscopic group ($P = 0.54$), and the average BMI in the open group was 29.72 kg/m² (5.03) versus 31.72 kg/m² (4.37) for the laparoscopic group ($P = 0.15$). Regarding the ASA score, while 29.4% of those who underwent an open approach were ASA 3 and 4 and only 15.8% of those who underwent a laparoscopic approach were ASA 3 and 4, no overall difference in the ASA score between the two groups ($P = 0.23$) [Table 1].

Baseline characteristics of the patients

BMI: Body mass index, ASA-PS: The American Society of Anesthesiologists Physical Status Classification System, SD: Standard deviation, COPD: Chronic obstructive pulmonary disease, CKD: Chronic kidney disease, HF: Heart failure, BA: Bronchial asthma, CT: Computerized tomography

Moreover, no statistically significant difference could be detected between the two groups in terms of the number of previous surgeries, comorbidities, laboratory results and maximum bowel diameter by CT finding.

Intraoperative data

Turning to the operative findings, the mean operative time in the laparoscopic approach was significantly lower than that in the open approach 71 min (16) versus 107 min (25), respectively ($P = 0.001$). In addition to that, a remarkable decrease in the mean blood loss could be detected in the case of the laparoscopic approach with 50 ml (33) versus 120 ml (74) in the open technique ($P = 0.001$). In addition to that, the open approach is associated with a higher incidence of iatrogenic injury (5.9% vs. 0%) that was managed with primary repair [Table 2].

Intraoperative data

SD: Standard deviation, SBO: Small bowel obstruction

Post-operative outcomes

Regarding the post-operative outcomes, the mean time to pass flatus was significantly lower in the laparoscopic group than in the open one 1.29 days (0.7) versus 2.78 days (1.3), respectively ($P = 0.001$). In addition to that, the length of post-operative hospital stay was higher in the open group than that in the laparoscopic group (3.8 days [2] and 2.39 days [1.6], respectively; $P = 0.001$). Moreover, a higher rate of early post-operative mortality was detected in the open group than that in the laparoscopic group (2 from 51 cases and 0 from 38 cases, respectively; $P = 0.001$). A higher rate of overall complication was found related to the open group than the laparoscopic group (39% vs. 10.5%, respectively) [Table 3].

Post-operative outcomes

AKI: Acute kidney injury, HF: Heart failure, SD: Standard deviation, SSI: Surgical site infection

DISCUSSION

In our study, laparoscopic adhesiolysis is associated with a remarkable decrease in operative time and blood loss in comparison to open adhesiolysis. In addition to that, those who underwent adhesiolysis by laparoscopy have a short hospital stay, early recovery and fewer post-operative complications. Moreover, open adhesiolysis is associated with a higher rate of early post-operative mortality. In addition to that, the incidence of iatrogenic injury was higher in the open group.

Since 1991, when Bastug et al.[13] reported the first laparoscopic adhesiolysis for SBO, the use of laparoscopy in case of adhesive bowel obstruction has increased dramatically. In 2008, according to data obtained from the National Inpatient Sample in the USA, Mancini et al.[14] stated that 11.3% of those who underwent an operation for adhesive bowel obstruction from 2002 to 2007 were done laparoscopically. In 2013, Kelly et al.[15] reported that 14.7% underwent laparoscopic adhesiolysis for SBO. In 2022, as reported by a single-centre retrospective study conducted in the USA, laparoscopic adhesiolysis reached 39% of the cases. In our study, 42.7% of the total number of cases were completed with the laparoscopic approach. This increase could be explained by the increase in the surgical experience of the surgeons, the availability and feasibility of the laparoscope and the possible better outcomes of laparoscopic adhesiolysis in comparison to the open approach that was reported by some studies.[16-18]

To begin with the operative time for the two approaches, according to Lombardo et al.[18] and Byrne et al.,[17] no statistically significant difference between the two procedures in terms of the operative time. In our study, a remarkable decrease in the length of the operation could be detected for laparoscopic adhesiolysis (71 min vs. 107 min, $P = 0.001$). These results are mirroring what was mentioned in some studies.[19,20] This decrease in the length of the operation was explained by Kelly et al.,[15] by the increase in the surgical experience. In addition to that, in line with what was reported by Byrne et al.,[17] intraoperative blood loss was significantly lower in the case of the laparoscopic approach.

Turning to the length of hospital stay, in agreement with several studies,[8,9,21] laparoscopic adhesiolysis is associated with a dramatic decrease in hospital stay in comparison to open adhesiolysis. Lin et al.[21] reported a mean hospital stay of 6.4 days versus 7.2 days, and according to Sallinen et al.,[9] the mean length of stay was 4.2 days versus 5.5 days. In addition to that, Chin et al.[8] reported a median length of stay of 8 days versus 13 days for laparoscopic versus open, respectively. In this study, the mean hospital stay was 2.4 days versus 3.8 days. On the other hand, on the contrary to Chin et al.,[8] in our study, the laparoscopic group has a quicker return to bowel function with the meantime to pass flatus 1.3 days versus 2.8 days for the laparoscopic versus open approaches.

According to previous studies,[9,16-18,20] the rate of early post-operative mortality ranged between 1.7% and 9% for open adhesiolysis and 0% and 5.7% for laparoscopic adhesiolysis. Furthermore, according to Chin et al.,[8] the mortality rate increases in the case of small bowel resection anastomosis. And as reported by a recent meta-analysis,[22] no significant difference in terms of the 30-day post-operative mortality between the two groups (relative risk = 0.70; 38 95% confidence interval = 0.14–3.51). However, in our study, the open approach was associated with a significantly higher mortality rate than the laparoscopic approach (3.9% vs. 0%). This result is conscious with what was reported by Kelly et al.[15] with a mortality rate of 4.7% versus 1.3% for open versus laparoscopic groups.

One of the disadvantages of laparoscopic adhesiolysis is the iatrogenic injury that may occur during trocar insertion, handling of the bowel and during adhesiolysis.[23,24] According to O' Connor and Winter,[23] 12 cases from 2005 who underwent laparoscopic adhesiolysis had trocar-induced bowel injury. Moreover, Wullstein and Gross[25] reported that iatrogenic injury was higher in the laparoscopic group than in the open group (26.9% vs. 13.5%) and all of these injuries occurred during adhesiolysis. In our study, no iatrogenic injury occurred by laparoscopic adhesiolysis. Our technique starts with the insertion of the Veress needle in the LUQ which gives us easy and rapid access to the peritoneal cavity, especially in obese patients, and minimises intraoperative gas leaks in comparison to the open technique. Then, we use an optical viewport to allow a safe entry to the abdominal cavity and avoid any bowel injury that may occur while inserting a trocar. Dissection of any adhesion to the abdominal wall gives us a safe view for inserting the second working port. Steep Trendelenburg patient position gives us greater access to the terminal ileum to begin handling the bowel from distal to proximal to avoid greater manipulation of the distended proximal bowel loops and avoid bowel injury. Using a cold scissor or Ligasure for adhesiolysis minimises thermal that may occur to the bowel. Thus, it can be said that laparoscopic management of SBO is a safe procedure and does not increase the incidence of enterostomy.

Turning to the overall incidence of complications, Nordin and Freedman[20] and Sallinen et al.[9] reported an equal overall rate of complications for the two approaches. However, Wullstein and Gross[25] and Kelly et al.[15] stated that the laparoscopic approach is associated with a significant decrease in the incidence of complications (19.2% vs. 26.4%, $P = 0.032$, and 10% vs. 22.2%, $P < 0.0001$). In addition to that, as reported by a meta-analysis conducted by Li et al.,[26] laparoscopic adhesiolysis is associated with a decrease in the overall incidence of complications (odds ratio = 0.42, $P < 0.01$). In this study, the incidence of complications was 10.5% for laparoscopy versus 39% for the open group ($P = 0.001$). In addition to that, a remarkable decrease in the rate of SSI could be detected in our study for the laparoscopic approach (2.6% versus 19.6%). This study mirrors Davies et al.,[19] which reported 0% SSI laparoscopic adhesiolysis versus 50% for the open approach. And, in line with the overall incidence of complications that was reported by Saleh et al.[16] (5.8% vs. 11.8%, $39 P < 0.001$). This decrease in the rate of SSI has been explained by Lin et al.[21] by the decrease in 30-day overall complications.

Sebastian-Valverde et al.[27] reported a conversion rate of 38.5%, and Dindo et al.[28] reported that 32.4% of the cases were converted into the open. The reasons for conversion were matted adhesions, inability to visualise obstruction, intraoperative complications and the need for resection. In our study, the incidence of conversion was 7%, and the main determining factor for conversion in our study was the experience of surgeons in advanced laparoscopy. This small number of conversions in our study could be explained by the exclusion of cases that needed resection that might have been converted into open after starting laparoscopically.

We have to admit the presence of several limitations in our study. First, it is a descriptive retrospective cohort study, and the selection of the patient to do either laparoscopic or open adhesiolysis depends on the surgeon which can cause a sort of selection bias. For these reasons, further randomised controlled trials are needed to address the presence of any selection bias. In addition to that, the small sample size in our study could lead to false-negative results in some analyses (type 2 errors). Furthermore, the short-term follow-up in our study was not enough to compare the long-term complication between the two groups (e.g. incisional hernia and recurrent adhesive SBO). However, we are aiming to continue the follow-up for the following 10 years. Moreover, the patient who underwent resection of the small bowel has been excluded. However, this was done to compare the outcomes and the rate of complications between laparoscopic and open approaches that could have been affected by including these patients.

Despite the presence of these limitations, we believe that the results obtained in our study are enough to confirm the advantage of using laparoscopy in simple adhesive SBO, especially if it is to be done by a skilled surgeon in the use of laparoscopy.

CONCLUSION

Laparoscopic adhesiolysis is a safe and feasible approach for the management of SBO and has better short-term outcomes, especially if done by skilled surgeons in advanced laparoscopic techniques.

Availability of data and materials

The dataset supporting the conclusions of this article is included within the article.

Authors' contributions

Beshoy Effat Elkomos contributed to data gathering, statistical analysis and writing the manuscript text. Karim Fahmy designed Figure 1 and Tables 1-3 and wrote the manuscript text. Kareem Ahmed Kamel wrote the main manuscript text.

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Conflicts of interest

There are no conflicts of interest.

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REFERENCES

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